

BHAVYA DAMANI

Data Analyst

bhavyadamani3125@gmail.com • +91 7045308570 • linkedin.com/in/bhavya-damani-1009b6277 • github.com/Bhavya031205 • Mumbai, India

PROFESSIONAL SUMMARY

Detail-oriented Data Analyst with practical experience across the complete analytics lifecycle — data ingestion, cleaning, exploratory analysis, SQL querying, statistical modelling, machine learning, and executive dashboarding in Power BI. Proficient in Python and SQL for building end-to-end analytics pipelines. Recognised at national level for applied predictive analytics research. Consistently delivers data-backed insights and actionable recommendations aligned to business objectives.

TECHNICAL SKILLS

Analytics & Visualisation: Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI, Tableau, Jupyter Notebook

Machine Learning: Scikit-learn, Logistic Regression, Random Forest, XGBoost, Predictive Modelling, Model Evaluation

Data & Databases: SQL, PostgreSQL, MongoDB, MySQL, ETL Pipelines, Data Cleaning, Feature Engineering

Statistical Methods: EDA, Hypothesis Testing, ANOVA, Correlation Analysis, Cohort Analysis, Regression

Languages: Python, SQL, JavaScript, Java, C/C++

Other Tools: Git, MS Excel (Intermediate), Power Query

PROJECTS

ClinicalIQ — Hospital Readmission Risk Analytics & Predictive Decision Support System • Python | Pandas | SQL | PostgreSQL | Scikit-learn | XGBoost | Power BI | Tableau

- Built a full clinical analytics pipeline on 85,000+ patient EHR records. Resolved 6,800+ nulls, corrected ICD coding schema inconsistencies, and engineered 11 features including Comorbidity Index Score and Prior Admission Count.
- EDA across 18+ visualisations revealed patients aged 65+ with 3+ prior admissions accounted for 61% of 30-day readmissions. Authored 14 SQL queries using CTEs and window functions for readmission rates, length-of-stay trends, and high-risk cohort segmentation.
- Compared 5 classification models; selected XGBoost (AUC-ROC 0.87, F1 0.81) with SMOTE for class imbalance. Delivered a 5-page interactive Power BI dashboard and four evidence-based clinical recommendations.

RetailIQ — Sales Intelligence & Predictive Analytics Platform • Python | Pandas | NumPy | SQL | PostgreSQL | Scikit-learn | Power BI

- End-to-end analytics pipeline on 120,000+ retail transaction records. Resolved 4,200+ nulls, removed duplicates, standardised date schemas, and engineered 6 features including CLV and Profit Margin %.
- EDA uncovered Technology category drove 42% of revenue from 18% of orders; Q4 ran 2.3x higher than Q1. Authored 12 SQL queries using window functions and CTEs. Quantified $r = -0.68$ correlation between discounting and margin; validated regional variance via ANOVA ($p < 0.01$).
- Trained Random Forest Regressor for quarterly forecasting (RMSE ~INR 4,200; 22% improvement over baseline). Built 4-page interactive Power BI executive dashboard with cross-filtering slicers.

ADHD Early Diagnostic Prediction System • Python | Scikit-learn | TensorFlow | Feature Engineering | Statistical Modelling

Runner-Up — Vidyavardhini's National Product Showcase (2nd Place)

- Preprocessed a neurological dataset covering behavioural assessments, physiological signals, and clinical indicators using class-balancing, normalisation, and recursive feature elimination.
- Compared Logistic Regression, Random Forest, Gradient Boosting, and a custom Deep Learning architecture; selected the optimal model using F1-score and AUC-ROC, weighting for the high cost of false negatives in a diagnostic setting. Awarded 2nd place at a national academic showcase.

EXPERIENCE

Data & Web Development Intern — Rotaract Club of Virar

Dec 2025 – Feb 2026

- Designed relational database schema in PostgreSQL to manage member records, events, and content; built reporting views and data feeds consumed by the frontend application

Web Development Intern — Vidyavardhini's College of Engineering & Technology

June 2024

- Delivered a responsive interface backed by MySQL; applied relational database fundamentals and UI/UX design principles

EDUCATION

B.E. — Artificial Intelligence & Data Science Engineering — *VCET, Palghar*

Aug 2023 – Present

CGPA: 8.0 / 10

HSC — Science — *SVKM's SVP College of Science*

Mar 2023

MHT-CET: 90.6%

ACHIEVEMENTS

- Runner-Up — Vidyavardhini's National Product Showcase for ADHD Predictive Diagnostic System (2nd place, institution-wide competition)
- CGPA 8.0 / 10 in a specialised AI & Data Science engineering programme